

FedHEP: Federated Hierarchical Ensemble Personalization

Single-Backbone 3-Tier Representation • Active-Class Logit Masking • Privacy-Preserving Sketch Clustering • Layered Subspace Byzantine Defense (SCCF + TTT)

EDGE CLIENT ON-DEVICE PIPELINE

Single Shared Backbone • No Model Duplication • 158.8 MB Peak VRAM (46.8% vs Ditto)

Local Batch

$$(X, Y) \sim \mathcal{D}_i$$

$$\mathcal{Y}_i \subset \{1 \dots C\}$$

Sparse Support

$$|\mathcal{Y}_i| \ll C \text{ (Non-IID)}$$

Shared Backbone

$$\Phi_{\theta}(x) \text{ (ResNet-9)}$$

Single forward pass
computed once

$$\text{Embedding } \mathbf{h} \in \mathbb{R}^d$$

1. Global Root Head $W_r \in \mathbb{R}^{d \times C}$

Consensus anchor $\rightarrow \mathbf{z}_r$ (Unmasked CE)

2. Cluster Parent Head $W_p \in \mathbb{R}^{d \times C}$

Collaborative pool $\rightarrow \mathbf{z}_p$ (ACLM Masked)

3. Private Local Head $W_l \in \mathbb{R}^{d \times C}$

Personalized head $\rightarrow \mathbf{z}_l$ (Strictly Private)

COMMUNICATION BUS

Synchronous Federated Rounds

Downlink Broadcast

Global Backbone: $\theta^{(t)}$

Global Root Head: $W_r^{(t)}$

Cluster Parent Head: $W_{p,k(i)}^{(t)}$

(Synchronized Peer Model)

Uplink Telemetry

Backbone Delta: $\Delta\theta_i$

Root Delta: $\Delta W_{r,i}$

Parent Delta: $\Delta W_{p,i}$

RP Sketch: $\mathbf{s}_i \in \mathbb{R}^{256}$

Active Support: $\mathcal{Y}_i \subset \{1 \dots C\}$

(Used for SCCF Defense)

PRIVACY BOUNDARY

Local Head $W_{l,i}$

NEVER UPLOADED
100% On-Device Isolation
Raw Data $\mathcal{D}_i$ Protected
Zero Telemetry Leakage

Safe against gradient inversion
and eavesdropping

FEDERATED SERVER ORCHESTRATOR

Layered Subspace Defense (SCCF + TTT) • Robust Consensus • Peer Clustering

Layered Subspace Byzantine Defense (SCCF + TTT)

Attacks Neutralized: Label Flip • Sign Flip • Gradient Ascent • Gaussian Noise

1. Subspace-Constrained Cosine Filtering (SCCF):

$$S_{ij}^{\text{SCCF}} = \frac{\langle \Delta\theta_i \odot \mathbf{M}_{ij}, \Delta\theta_j \odot \mathbf{M}_{ij} \rangle}{\|\Delta\theta_i \odot \mathbf{M}_{ij}\|_2 \|\Delta\theta_j \odot \mathbf{M}_{ij}\|_2}, \quad \mathbf{M}_{ij} = \mathbb{I}(\mathcal{Y}_i \cap \mathcal{Y}_j \neq \emptyset)$$

- Projects cosine similarity strictly onto observed label overlap subspace
- Prevents false rejection of honest specialized clients under skew!**

2. Adaptive Norm Bounding & Temporal Trust Tracking (TTT):

$$\|\Delta\theta_i\|_2 \leq \tau_{\text{clip}}, \quad \tau_i^{(t)} = \beta_{\tau} \tau_i^{(t-1)} + (1 - \beta_{\tau}) s_i^{(t)}$$

Multi-round historical momentum isolates insidious poisoning over time

Robust Consensus Aggregation Engine

$$\theta^{(t+1)} \leftarrow \theta^{(t)} + \sum_{i \in \mathcal{H}} \tilde{\tau}_i \Delta\theta_i, \quad W_r^{(t+1)} \leftarrow W_r^{(t)} + \sum_{i \in \mathcal{H}} \tilde{\tau}_i \Delta W_{r,i}$$

- Trust-calibrated weighting $\tilde{\tau}_i$ shields global parameters from poisoning
- Preserves high consensus representations on homogeneous/IID distributions
- Verified benchmark: 72.73% IID accuracy on CIFAR-10 (40.94% on CIFAR-100)

Privacy-Preserving Collaborative Peer Clustering

1. Sketch Cosine Affinity Routing:

Assign client $i \rightarrow C_k \iff k = \arg \max_j \cos(\mathbf{s}_i, \mu_j)$

2. Momentum-Stabilized Cluster Parent Head Aggregation:

$$W_{p,k}^{(t+1)} \leftarrow \beta_c W_{p,k}^{(t)} + (1 - \beta_c) \bar{W}_{p,k}, \quad \bar{W}_{p,k} = \sum_{i \in C_k} \frac{w_i}{N_k} (\Delta W_{p,i} + W_{p,k}^{(t)})$$

Cluster 1

Head $W_{p,1}$

Cohort C_1

Cluster 2

Head $W_{p,2}$

Cohort C_2

Cluster 3

Head $W_{p,3}$

Cohort C_3

Temperature-Calibrated Ensemble Inference:

$$\mathbf{z}_{\text{ens}} = w_l \frac{\mathbf{z}_{l,i}}{T_l} + w_p \frac{\mathbf{z}_{p,i}}{T_p} + w_r \frac{\mathbf{z}_{r,i}}{T_r} \implies \hat{y} = \arg \max_{c \in \mathcal{Y}} \mathbf{z}_{\text{ens},c}$$

Preserves prediction sharpness and eliminates probability dilution without dual-model memory overhead

Active-Class Logit Masking (ACLM)

Eliminates Negative Gradient Interference:

$$\tilde{z}_{i,c} = z_{i,c} \text{ (if } c \in \mathcal{Y}_i), \quad \tilde{z}_{i,c} = -\infty \text{ (if } c \notin \mathcal{Y}_i)$$

$$\implies \frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_{i,c}} = 0 \quad (\forall c \notin \mathcal{Y}_i)$$

- Shields shared backbone Φ_{θ} from unobserved penalty
- Eliminates gradient corruption under extreme skew

Skew-Calibrated Bernstein Loss Weighting:

$$R_{\text{skew}}^{(i)} = 1 - \frac{H(\mathbf{p}_i)}{\ln C} \in [0, 1] \quad (\text{Entropy Skew})$$

$$\lambda_r = a_i + (1 - a_i)(R_{\text{skew}}^{(i)})^2, \quad \lambda_p = 2R_{\text{skew}}^{(i)}(1 - R_{\text{skew}}^{(i)})$$

$$\lambda_l = (1 - R_{\text{skew}}^{(i)})^2, \quad a_i = \max\left(\frac{1}{2K}, \frac{|\mathcal{Y}_i|}{C}\right)$$

Composite Local Training Objective:

$$\mathcal{L}_{\text{total}} = \alpha_r \mathcal{L}_r + \alpha_p \mathcal{L}_p^{\text{mask}} + \alpha_l \mathcal{L}_l^{\text{mask}}$$

Smooth partition-of-unity dynamic adaptation

Random Projection (RP)

Privacy-Preserving Sketching:

- Parent Delta: $\Delta W_{p,i} \in \mathbb{R}^{d \times C}$

Shared Gaussian Matrix:

$$\mathbf{R} \in \mathbb{R}^{256 \times (d \cdot C)}, \quad R_{jk} \sim \mathcal{N}(0, 1/m)$$

Low-Dim Normalized Sketch:

$$\mathbf{s}_i = \frac{\mathbf{R} \Delta W_{p,i}}{\|\mathbf{R} \Delta W_{p,i}\|_2} \in \mathbb{R}^{256}$$

- Information-Theoretic Privacy:**
Underdetermined system ($m \ll D$)
prevents raw parameter inversion

Ultra-Lean Transmission:

Payload: 1.0 KB / round (10× comp.)